a

HUMAN PROTEOMICS

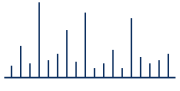


20,431
reference proteins



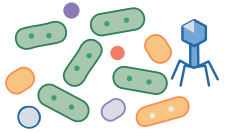
Human reference proteome

Database search



Identified proteins

CONVENTIONAL METAPROTEOMICS



Matched metagenome	Catalog reduction	Iterative search
DNA	Large catalog	Large catalog
Sequencing		First search
Assembly		Selected proteins or genomes
Gene prediction	Selection or clustering	Second search

Database search



Identified proteins

FastaLake

MS/MS



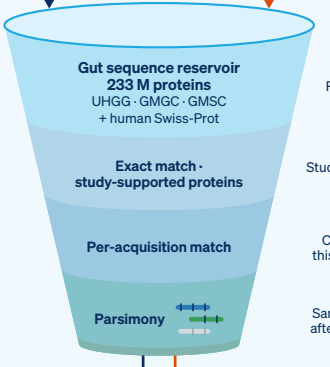
De novo

AlphaNovo peptide predictions
24,560 unique peptides



Composition-matched scrambled peptides
36,588 unique peptides





	Real	Scrambled
Proteins in reservoir	233 M	233 M
Study-supported proteins	18 M	5,841
Candidates, this acquisition	6.67 M	5,841
Sample FASTA after parsimony	12,606	525

Database search

29,663 peptides



Accepted peptides (authentic)

0 peptides



Accepted peptides (scrambled)